

ImageNet-R Stage-31 Macro-Token Convergence Audit

Clean-only optimization audit - stage 31 - locked test unopened

Main result

The predeclared selection rule minimized clean validation NLL. It chose effective batch 64 and peak learning rate $3e-05$. At each seed's minimum-NLL checkpoint, this schedule reached 74.341% mean validation accuracy and 1.0929 mean NLL (1993: 74.483%, 1994: 74.549%, 1995: 73.991%). The same-split joint-IID control reached 77.862% accuracy and 0.9425 NLL after its fixed fifth epoch. The selected macro mean therefore differed from joint IID by -3.520 accuracy points and +0.1504 NLL.

Checkpoint policy changes the accuracy diagnosis. The selected schedule's per-seed maximum accuracy averaged 76.233% (1993: 76.451%, 1994: 75.894%, 1995: 76.353%), still -1.629 points from joint IID. In the exploratory nine-cell screen, batch 128 with peak learning rate 0.0003 reached 78.091% at epoch 33, +0.230 points relative to joint IID, but its NLL was 1.1902. That one-seed, post-screen maximum was not replicated and is not the selected estimate.

The exact legacy rerun reached 73.762% at its minimum-NLL checkpoint (epoch 2). The selected screening cell changed seed-1993 accuracy at minimum NLL by +0.722 points. Every selected run peaked before epoch 50; more epochs under the same schedule are not supported by validation NLL.

What was tested

All macro models used the same one-block 12,055,496-parameter architecture and the same frozen stage-31 hierarchy representations. Seed 1993 crossed effective batches 64, 128, and 512 with peak AdamW learning rates $3e-5$, $1e-4$, and $3e-4$. Each cell used a five-percent linear warmup followed by cosine decay through epoch 50. The legacy control used constant $3e-4$ for 20 epochs. The selected schedule was repeated with seeds 1994 and 1995.

The joint-IID control used the identical 12,194 fit and 3,049 validation identities, but it trained a fresh rank-16 QKV-plus-fc1 LoRA and affine 124-class head for five epochs. It therefore measures what joint feature adaptation can achieve on this clean split; it is not a gate.

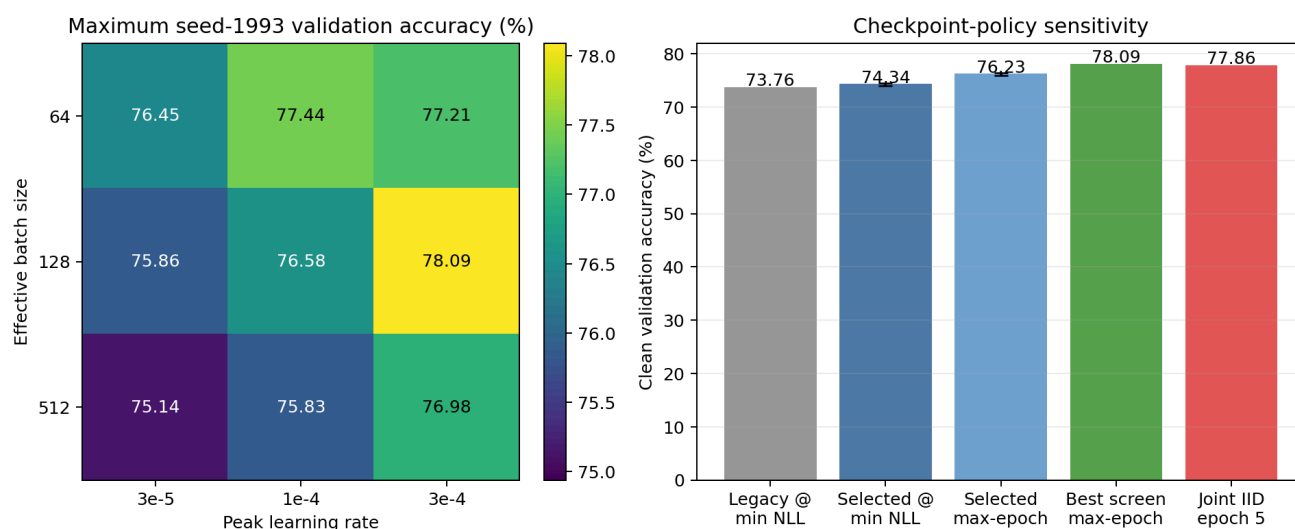


Figure 1. Left: maximum seed-1993 validation accuracy over 50 epochs. Right: accuracy at the predeclared minimum-NLL checkpoints, per-seed maximum accuracy for the selected schedule, the unreplicated best screened maximum, and the fixed joint-IID endpoint. Macro error bars show observed three-seed ranges, not confidence intervals.

Interpretation

The macro models reached near-perfect fit accuracy, while every selected seed minimized validation NLL by epoch 9. More optimizer work therefore does not close the probabilistic gap: after those checkpoints, validation NLL rises even when top-1 accuracy sometimes improves. The exploratory 78.091% maximum shows that the fixed macro inputs can support a competitive decision boundary on this development split, but its much worse NLL and lack of replication point to calibration and generalization instability rather than simple non-convergence.

The comparison cannot isolate classifier architecture from representation quality. Joint IID adapts one LoRA jointly over all 124 classes, while the macro classifier receives frozen node-specific features. The joint model's 0.9425 validation NLL, achieved before its fit accuracy saturated, is evidence that joint feature adaptation produces a cleaner representation than merely fitting the macro classifier longer.

The experiment never requested a test image. These are development measurements only and do not revise the locked-test result from v8.

Full optimizer matrix

Batch	Peak LR	Min-NLL epoch	Acc. @ min NLL	Min NLL	Max-acc. epoch	Max acc.	NLL @ max acc.
64	3e-05	7	74.483	1.0903	47	76.451	1.1905
64	0.0001	5	74.287	1.1209	14	77.435	1.1328
64	0.0003	2	71.105	1.2329	45	77.206	1.3116
128	3e-05	11	75.139	1.0935	25	75.861	1.1318
128	0.0001	5	73.926	1.1068	20	76.582	1.1465
128	0.0003	28	77.829	1.1852	33	78.091	1.1902
512	3e-05	22	74.582	1.0982	41	75.139	1.1071
512	0.0001	8	74.254	1.0925	42	75.828	1.1441
512	0.0003	14	76.648	1.1199	22	76.976	1.1351

Stage-31 macro-token clean learning curves

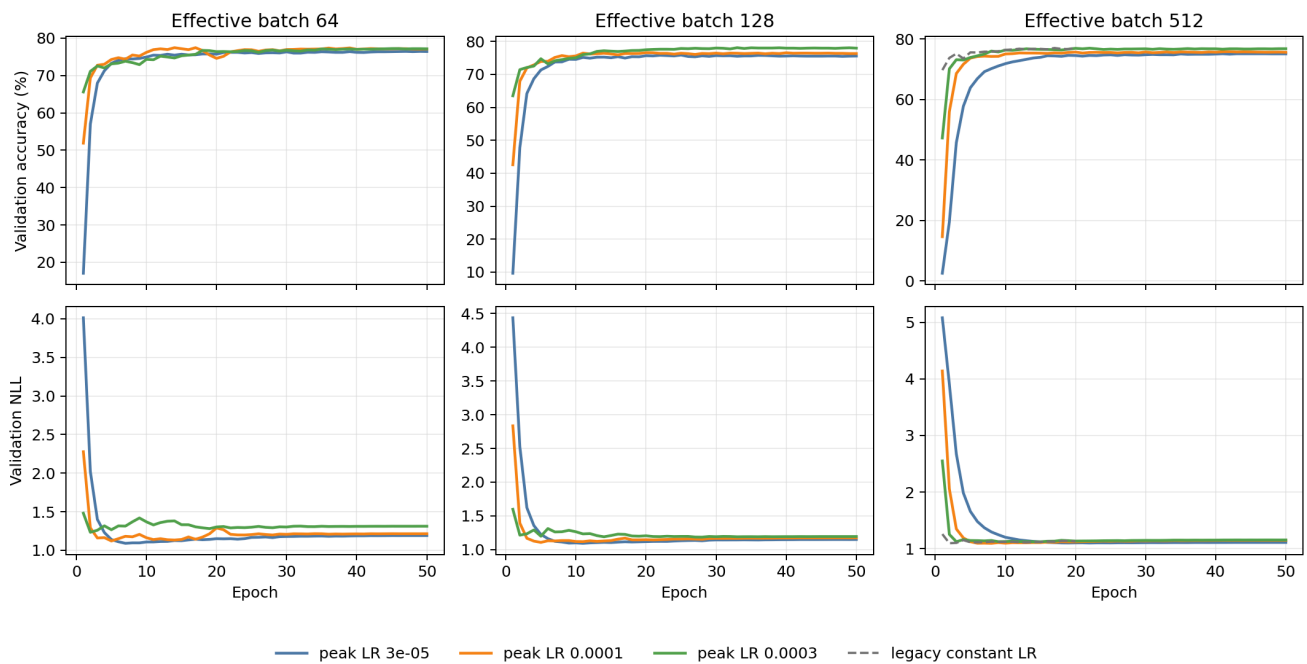


Figure 2. Validation accuracy and negative log likelihood (NLL) through every scheduled epoch. The dashed legacy curve appears only in the effective-batch-512 panel.

Update-count view

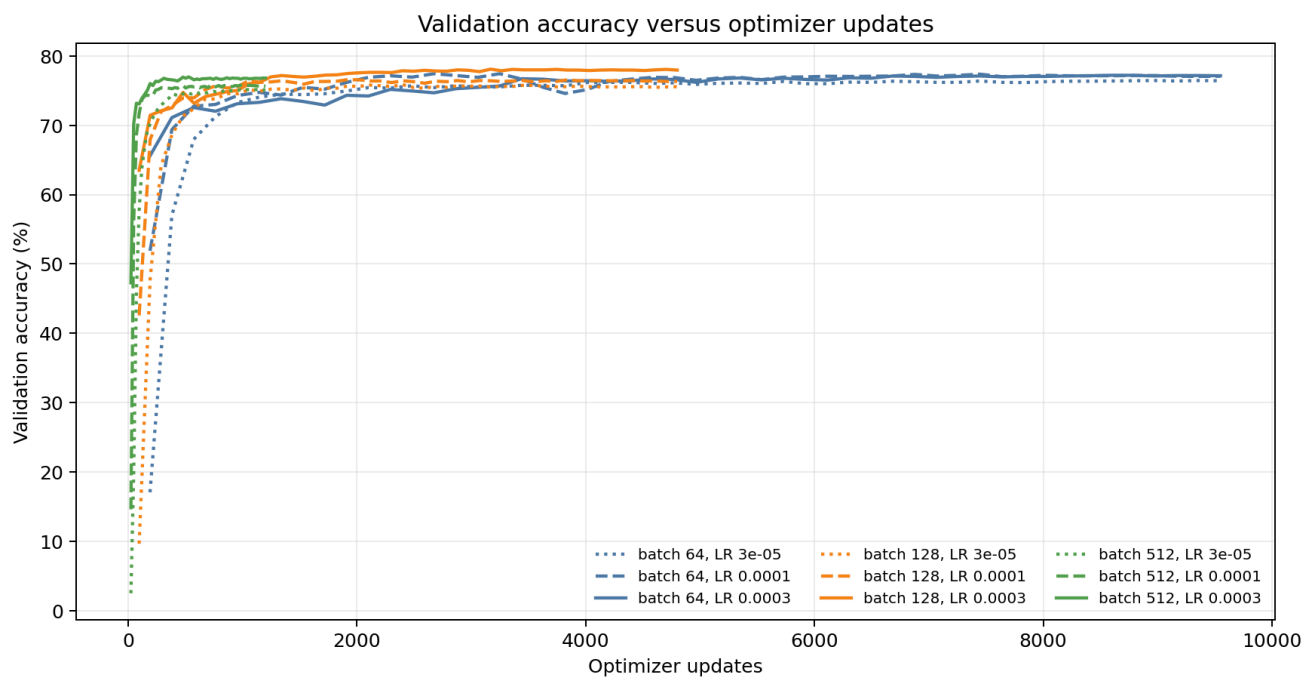


Figure 3. The same validation measurements indexed by optimizer updates rather than epochs.

Reproducibility

The source v8 run, fit-only hierarchy, split, environment, code, and resolved configuration are content-addressed. Epoch rows were hash-chained and fsynced before the next epoch. An immediate cache replay used 76,215 cached node-example rows, performed zero adapted-token forwards, and preserved both population identities.